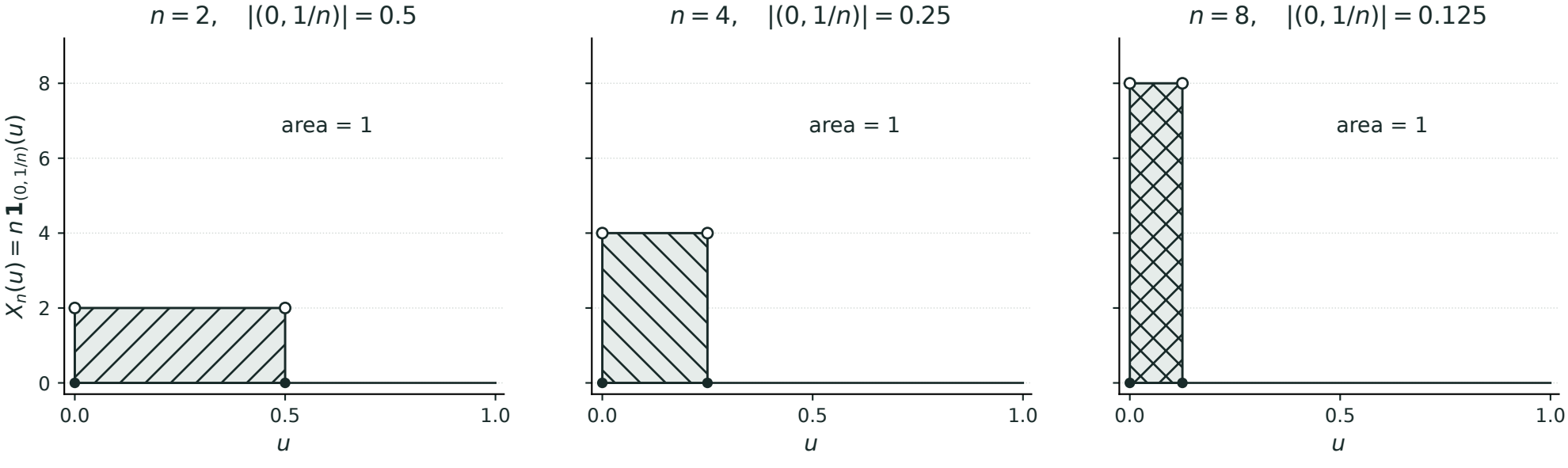
Height grows, support shrinks, area stays one
$n = 2, \quad |(0, 1/n)| = 0.5$
$n = 4, \quad |(0, 1/n)| = 0.25$
$n = 8, \quad |(0, 1/n)| = 0.125$
area = 1
area = 1
area = 1
$X_n(u) = n\,\mathbf{1}_{(0,\,1/n)}(u)$
0
2
4
6
8
0.0
0.5
1.0
u
$X_n \to 0$ a.e., but $\mathbb{E}X_n = 1$. For every $K \geq 0$: $\sup_n \mathbb{E}[X_n \mathbf{1}_{\{X_n > K\}}] = 1$; UI fails.